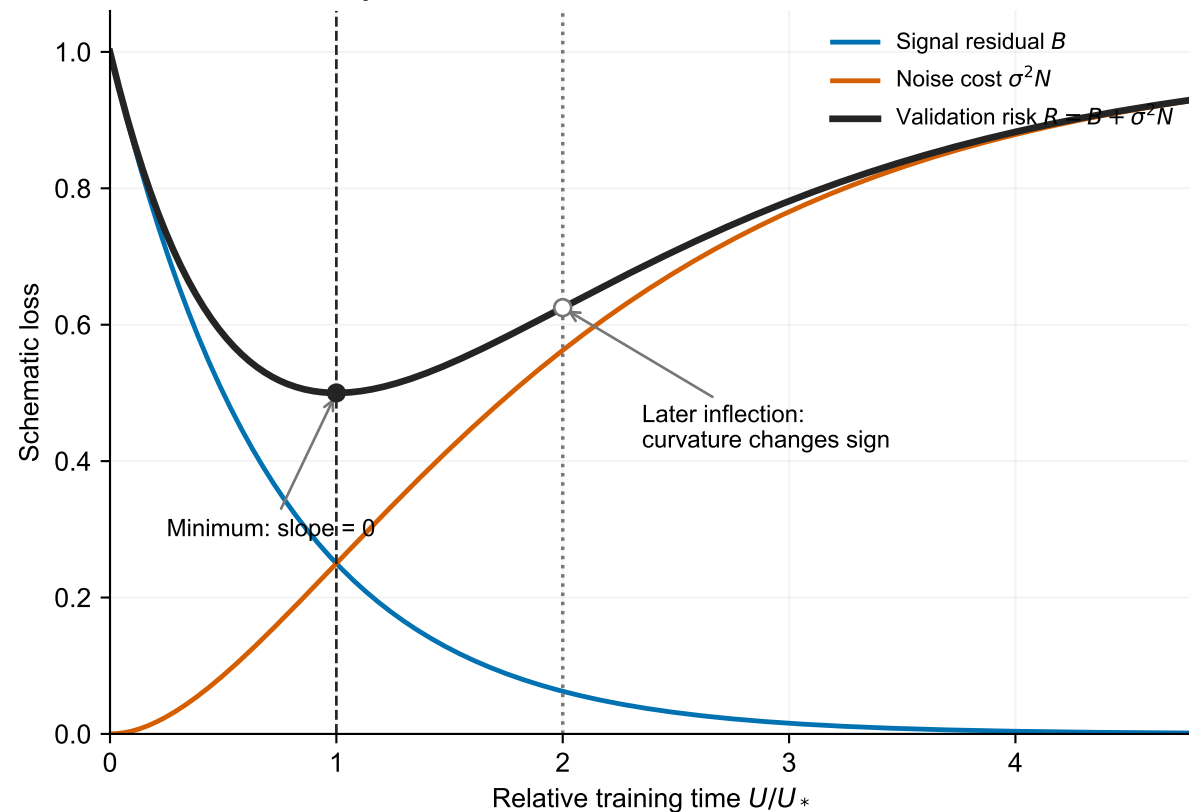
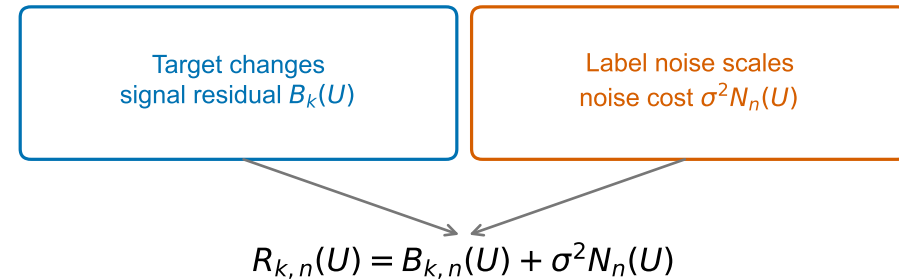


A one-mode example of the mechanism



At fixed data and features



Before the minimum

$-B' > \sigma^2 N'$: risk falls

At the minimum

$-B' = \sigma^2 N'$: rates balance

After the minimum

$-B' < \sigma^2 N'$: risk rises

A shared noise shape does not imply a universal rise-time exponent.

Noise fitting begins before the rise. The minimum is a rate balance; it need not coincide with train loss approaching zero.

Schematic only: $B = e^{-4\lambda U}$, noise = $(1 - e^{-2\lambda U})^2$, $U_* = (\log 2)/(2\lambda)$. Real curves can have several inflections.